

# **Design and Characterization of Ag/SmFeO<sub>3</sub> Perovskite Nanostructures with improved Electrochemical Properties**

Submitted in partial fulfillment of the requirements for the award of the degree of

## **BACHELOR OF SCIENCE IN PHYSICS**

by

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## **Dr. N.G.P. ARTS AND SCIENCE COLLEGE**

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Approved by Government of Tamil Nadu & Accredited by NAAC with A<sup>++</sup> Grade (3<sup>rd</sup> Cycle.3.64 CGPA)

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## DECLARATION

I, **Mr. KISHORE R (Reg. No.: 231PH010)** hereby declare that the project report entitled **“Design and Characterization of Ag/SmFeO<sub>3</sub> Perovskite Nanostructures with improved Electrochemical Properties”** submitted in partial fulfilment of the requirement for the award of the degree of **BACHELOR OF SCIENCE IN PHYSICS** is a record of original work done during the period of 2023-26 under the supervision and guidance of **Dr. K. Girija, M.Sc., M.Phil., Ph.D., Associate Professor and Head**, Department of Physics, Dr. N.G.P. Arts and Science College, Coimbatore - 48, and it has not formed on the basis of award of any Degree/ Diploma/ Associateship/ Fellowship or other similar title to any candidate of any university.

**(Mr. KISHORE R)**

**Reg. No.: 231PH010**

**Place:** Coimbatore

**Date:**

## **CERTIFICATE**

This is to certify that the project report entitled “**Design and Characterization of Ag/SmFeO<sub>3</sub> Perovskite Nanostructures with improved Electrochemical Properties**” submitted in partial fulfilment of the requirement for the award of the degree of **BACHELOR OF SCIENCE IN PHYSICS** is a record of original work done by **Mr. KISHORE R (Reg. No.: 231PH010)** during the period (2025-2026) of his study in the **Department of Physics, Dr. N.G.P. Arts and Science College, Coimbatore-48** under my supervision and guidance, and the internship report has not formed the basis for the award of any Degree/ Diploma/ Associateship / Fellowship or other similar title to any candidate of any university.

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**Date :**

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**External Examiner**

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**(KISHORE R)**

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# CHAPTER I

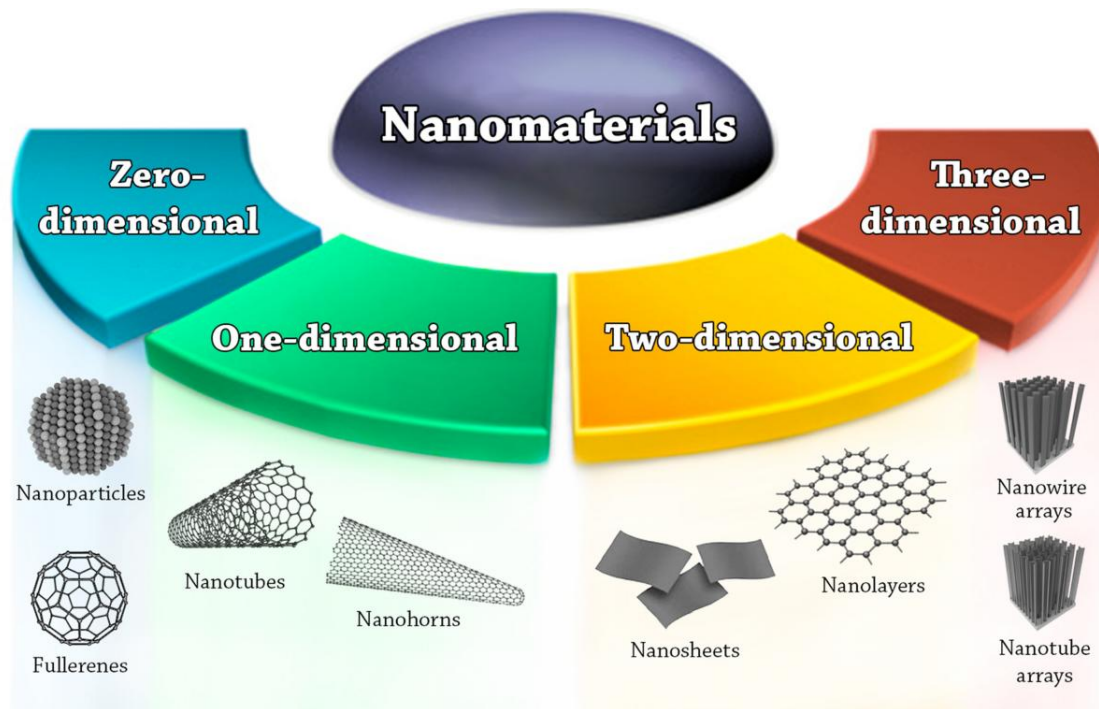
## INTRODUCTION

### 1.1 NANOSCIENCE AND NANOTECHNOLOGY

Nanoscience is the study of materials and phenomena that occur at the nanometer scale, typically between 1 and 100 nanometers. At this scale, materials exhibit unique physical, chemical, optical, electrical, and magnetic properties that are significantly different from their bulk counterparts. The prefix “nano” is derived from the Greek word meaning dwarf, representing one-billionth ( $10^{-9}$ ) of a meter. Understanding matter at this scale allows scientists to manipulate atoms and molecules directly. Nanotechnology, on the other hand, refers to the practical application of nanoscience. It involves designing, synthesizing, characterizing, and applying materials and devices at the nanoscale. This interdisciplinary field integrates physics, chemistry, biology, materials science, and engineering. The goal is to develop advanced materials with enhanced functionality and performance. At the nanoscale, surface effects become dominant due to the high surface-to-volume ratio. This results in increased chemical reactivity and altered electronic behaviour. Quantum mechanical effects also become significant, leading to phenomena such as quantum confinement. These changes make nanomaterials extremely useful in electronics, medicine, catalysis, and energy applications.[1]

Nanotechnology has revolutionized several industries including electronics, healthcare, environmental remediation, and renewable energy. For example, nanoscale semiconductors are used in advanced solar cells and sensors. Nanoparticles are used in drug delivery systems to target specific cells in the human body. The development of nanotechnology tools such as scanning electron microscopy (SEM), transmission electron microscopy (TEM), atomic force microscopy (AFM), and X-ray diffraction (XRD) has enabled precise characterization of nanomaterials.[2] These techniques help in understanding structural and morphological properties at atomic resolution. Nanoscience and nanotechnology are considered key technologies of the 21st century. Their impact is visible in next-generation materials, smart devices, and sustainable energy solutions. Continuous research in this field is paving the way for innovative breakthroughs in science and engineering.[3]

## 1.2 NANOMATERIAL

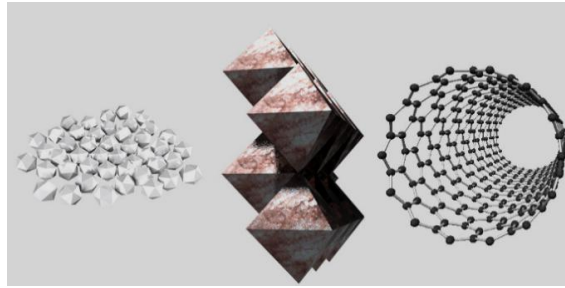


**Fig 1.1:** Nanomaterials and Classification

Nanomaterial is a heart of nanoscience and nanotechnology. In recent years nanotechnology has become one of the most important and exciting head fields in physics, biology and engineering. It shows great ability for providing us in the near future with many developments that will change the direction of technological advances in a wide range of applications. Nanotechnology is based on nanomaterial, size in nanometre (nm) range. Actually, the size of the nanomaterial is less than the size of 100 nanometres (a nanometre is one millionth of a millimetre nearly 100,000 times smaller than the diameter of a human hair). The particle which are smaller than the characteristic lengths associated with particular phenomena frequently show new chemistry and physics, important to new manners which depends on the size. For example, structure of electronic, reactivity, melting temperature, conductivity and mechanical properties have all been observed to change when the particles become smaller than a critical size.[4]

Nanomaterials have emerged as one of the most significant advancements in materials science, offering exceptional properties due to their nanoscale dimensions. These materials, typically defined as having at least one dimension in the range of 1-100 nanometres, exhibit unique chemical, mechanical, electrical, and optical characteristics that distinguish them from their bulk counterparts. The development and application of nanomaterials have paved the way

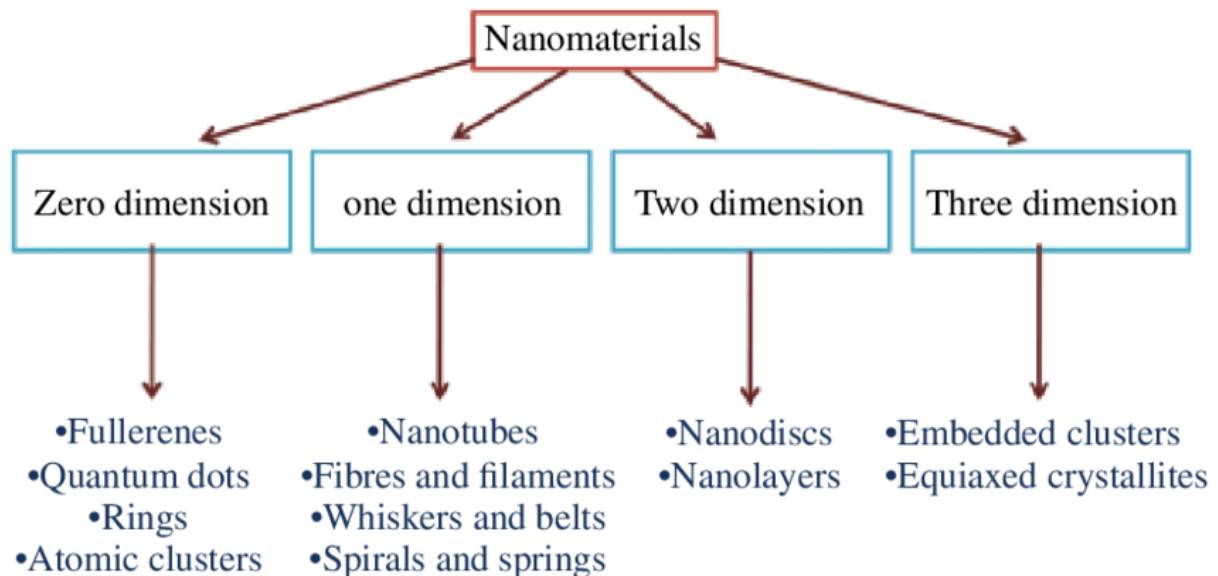
for innovations in various fields, including medicine, electronics, energy, and environmental sciences.[5]



**Fig 1.1.1** Framework for Sustainable Nanomaterial

### 1.2.1 CLASSIFICATION OF NANOMATERIALS

Nanomaterials can be classified into different categories based on their composition and structure. Broadly, they can be categorized into four types:

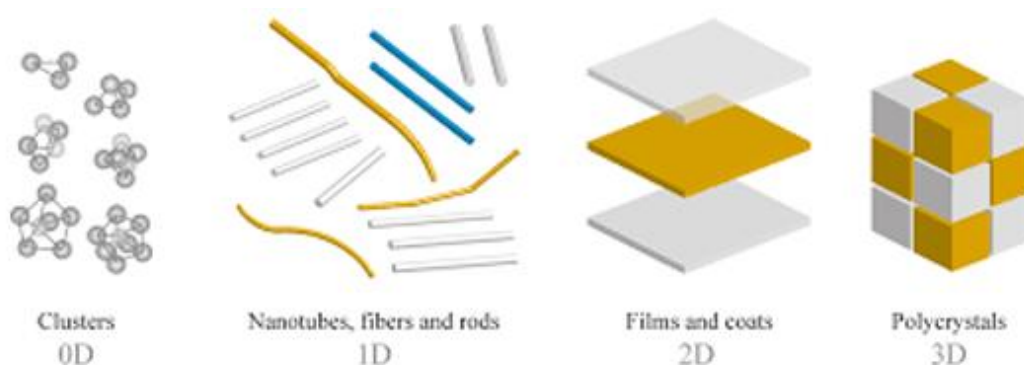


**Fig 1.1.2** Classification of Nanomaterials

### 1.2.2 DIMENSIONAL CLASSIFICATION OF NANOMATERIAL

1. **Zero-Dimensional (0D)** – Nanoparticles, quantum dots (all dimensions in nanoscale).
2. **One-Dimensional (1D)** – Nanotubes, nanowires, nanorods (one dimension outside nanoscale).
3. **Two-Dimensional (2D)** – Graphene, nanofilms, nanosheets (two dimensions outside nanoscale).

4. **Three-Dimensional (3D)** – Nanocomposites, nanoporous materials (all dimensions outside nanoscale but with nanostructures).



**Fig 1.2** Classification of Nanomaterials

### 1.3 NANOPARTICLES

Nanoparticles are particles whose size lies in the nanometer range, typically between 1–100 nm. Due to their small size, nanoparticles possess unique physical and chemical properties compared to bulk materials. The reduction in size increases surface area significantly, which enhances their reactivity and interaction with the surrounding environment. The most important feature of nanoparticles is their high surface-to-volume ratio. A large fraction of atoms exists on the surface rather than inside the material. This leads to improved catalytic activity, enhanced adsorption properties, and better interaction with light and electromagnetic radiation. Nanoparticles can be composed of metals, metal oxides, semiconductors, polymers, or composite materials. Examples include silver nanoparticles, iron oxide nanoparticles, zinc oxide nanoparticles, and ferrite-based nanoparticles. Each type exhibits distinct properties suitable for specific applications.[6] Another key feature of nanoparticles is quantum confinement. When the particle size becomes comparable to the de Broglie wavelength of electrons, discrete energy levels are formed instead of continuous bands. This affects electrical conductivity and optical absorption. Nanoparticles are widely used in medicine for drug delivery and imaging. In environmental science, they are used for pollutant removal and water purification. In energy research, they play a crucial role in photocatalysis and water splitting applications. However, nanoparticles also raise concerns regarding toxicity and environmental impact. Therefore, proper synthesis, stabilization, and handling are important aspects of nanoparticle research.[7]

## 1.4 ORIGIN OF NANO

Many nano forms of substance present around us. One of the initial nano sized objects is known to us was made of gold. Faraday prepared colloidal gold in 1856 and called it divided metals. In his diary dated 2 April 1856, Faraday called the particle he made the ‘divided state of gold’. The solutions he prepared are preserved in the Royal institution. Metallic gold, when divided in to fine particle ranging from size of 10 - 500 nm particles, can be suspended in water. In 1890, the German bacteriologist Robert Koch found that compounds made with gold repressed the growth of bacteria. He won the Nobel Prize for medicine in 1905. The habit of gold in medical preparations is not new. In Indian medical system called Ayurveda, gold is used in several preparations. ‘Saraswatharishtam’ called one popular preparation, good for memory development. In Alexandria, alchemists developed an influential colloidal medicine known as ‘liquid gold’, a preparation that was meant to reestablish youth. The great Experimenter and founder of modern medicine developed many highly successful treatments from metallic minerals including gold. On December 29, in 1959, the Nobel Prize winning physicist, Richard Feynman give a talk at the annual meeting of the American physical society entitled “*There’s plenty of room at the bottom*”. [8] The recent development of technology shows that reductions are needed in the dimensions of devices and active materials. This is a proof in case of computer technology. The number of transistors used in an IC (integrated circuit) has increased in the past 40 years. In 1965, Gordon Moore, the co-founder of Intel, detected that the number of transistors per square inch on integrated circuits doubled every year. Moor expected that this style would continue likely in future. [9]

### 1.4.1 SIZE DEPENDENT PROPERTIES

When the size of a material is reduced to the nanometer scale, its physical and chemical properties no longer remain the same as those of the bulk material. This phenomenon is known as size-dependent behavior. At the nanoscale, the number of atoms present on the surface becomes comparable to those inside the material. As a result, surface energy increases significantly, and this strongly influences the overall behavior of the material. The dominance of surface atoms modifies bonding characteristics, lattice parameters, and thermodynamic stability. [10] One of the most important reasons for size-dependent properties is quantum confinement. In bulk materials, electrons can move freely within continuous energy bands. However, when particle size approaches the de Broglie wavelength of electrons, their motion becomes confined. This confinement leads to discrete energy levels instead of continuous

bands, altering the band gap energy. As the particle size decreases, the band gap generally increases, which directly affects electrical conductivity and optical absorption properties.

Another major factor contributing to size-dependent properties is the increased surface-to-volume ratio. In nanoparticles, a large fraction of atoms lies at or near the surface. These surface atoms have unsatisfied bonds and higher reactivity compared to interior atoms. This enhances catalytic activity, adsorption capacity, and chemical reactivity. In photocatalytic materials such as cerium orthoferrite ( $\text{SmFeO}_3$ ), smaller particle size can improve light absorption efficiency and increase the number of active sites for water splitting reactions.[11]

### **1.4.2 ELECTRICAL PROPERTIES**

At the nanoscale, electron motion becomes confined within a small region. In bulk materials, electrons move freely within continuous energy bands. However, when particle size becomes comparable to the electron wavelength, quantum confinement occurs. Due to this, the energy levels become discrete rather than continuous. As the particle size decreases, the band gap energy generally increases in semiconductor nanoparticles. This means smaller particles require higher energy for electron excitation. Because of this increase in band gap, electrical conductivity decreases in some materials, while in certain nanowires or nanotubes, conductivity may improve due to reduced scattering. Surface defects and dangling bonds also influence charge carrier mobility. Trapping of electrons and holes at the surface modifies electrical transport behavior. In photocatalytic materials like cerium orthoferrite ( $\text{SmFeO}_3$ ), reduced particle size enhances charge separation efficiency, which is very important for water splitting applications. Thus, electrical properties are highly tunable by controlling particle size during synthesis.[12]

### **1.4.3 OPTICAL PROPERTIES**

Optical behavior of nanomaterials strongly depends on size due to quantum confinement. In semiconductor nanoparticles, as size decreases, the absorption edge shifts toward shorter wavelengths. This phenomenon is called blue shift. It occurs because of the increase in band gap energy. In metallic nanoparticles such as gold or silver, a unique phenomenon called surface plasmon resonance (SPR) occurs. Here, conduction electrons collectively oscillate when light interacts with the nanoparticle. The wavelength of this resonance depends on particle size, shape, and surrounding medium. Smaller nanoparticles often show enhanced light absorption and emission properties. This is beneficial in applications

such as solar cells, photocatalysis, LEDs, and sensors. For water splitting applications, better light absorption increases photocatalytic efficiency. Therefore, optical properties can be engineered by adjusting nanoparticle size and morphology.[13]

#### **1.4.4 MAGNETIC PROPERTIES**

Magnetic behavior changes significantly when particle size decreases. In bulk materials, magnetic domains exist to minimize magnetic energy. However, when particle size becomes smaller than a critical diameter, the material becomes a single-domain particle. Further reduction in size leads to superparamagnetism. In this state, thermal energy becomes greater than magnetic anisotropy energy, and magnetization randomly flips direction without an external magnetic field. As a result, there is no hysteresis loss, zero coercivity, and zero remanence. This property is highly useful in biomedical applications, magnetic storage, and sensors. In ferrite-based nanoparticles, controlling size allows tuning of magnetic saturation and coercivity values. Thus, magnetic properties are extremely sensitive to particle size variation.[14]

#### **1.4.5 THERMODYNAMIC PROPERTIES**

Thermodynamic properties such as melting point, phase transition temperature, and heat capacity also change at nanoscale. Due to the high surface energy of nanoparticles, melting point decreases compared to bulk materials. This phenomenon is known as melting point depression. Smaller particles have weaker atomic bonding because many atoms are located at the surface. As a result, less energy is required to break bonds during melting. Phase transitions may occur at lower temperatures. Diffusion rates are also higher in nanoparticles because atoms can move more easily along grain boundaries. These changes affect synthesis processes, sintering behavior, and stability of nanomaterials. Therefore, thermodynamic stability depends strongly on particle size.[15]

#### **1.4.6 MECHANICAL PROPERTIES**

Mechanical strength and hardness often improve when materials are reduced to nanoscale. This is explained by the Hall–Petch relationship, which states that decreasing grain size increases material strength due to grain boundary strengthening. Nanomaterials contain a high density of grain boundaries that block dislocation motion. This results in increased hardness and yield strength. However, when grain size becomes extremely small, inverse Hall–

Petch behavior may occur, where strength decreases due to grain boundary sliding. Elastic modulus and fracture toughness may also change at nanoscale. These enhanced mechanical properties are useful in nanocomposites, coatings, and structural materials.[16] Hence, particle size plays a crucial role in determining mechanical performance.

#### **1.4.7 CHEMICAL REACTIVITY**

Chemical reactivity increases significantly as particle size decreases. Because of the large surface-to-volume ratio, more atoms are available for reaction. Surface atoms have unsatisfied bonds, making them highly reactive. This increased reactivity enhances catalytic efficiency. In photocatalysts like  $\text{SmFeO}_3$  nanoparticles, smaller size increases active surface sites, leading to improved water splitting efficiency. Adsorption capacity also increases in nanoparticles due to higher surface area. This is useful in environmental remediation and gas sensing applications. Thus, chemical activity is strongly size-dependent [17].

#### **1.5 NANOPARTICLE SYNTHESIS METHODS**

The synthesis of nanoparticles is a crucial step in nanoscience research because the physical, chemical, optical, and magnetic properties of nanomaterials strongly depend on their size, shape, crystallinity, and surface structure. Controlling these parameters during synthesis determines the final performance of the material in applications such as catalysis, sensing, magnetic storage, and water splitting. Nanoparticle synthesis methods are broadly classified into two main approaches: top-down and bottom-up techniques. In the top-down approach, bulk materials are broken down into nanoscale particles using mechanical or physical methods. In contrast, the bottom-up approach builds nanoparticles atom-by-atom or molecule-by-molecule through chemical reactions. The choice of synthesis method depends on several factors such as desired particle size, uniformity, cost, scalability, crystallinity, and application. [18] For example, photocatalytic applications require high crystallinity and controlled morphology, while biomedical applications require surface functionalization and stability. Another important consideration in nanoparticle synthesis is the prevention of agglomeration. Due to high surface energy, nanoparticles tend to cluster together. Stabilizing agents or surfactants are often used to control particle growth and maintain dispersion. Synthesis methods also influence defect concentration, surface states, and band gap energy. These factors are especially important in semiconductor and perovskite-based nanoparticles used in energy

applications. Below is a detailed classification and explanation of major nanoparticle synthesis methods.[19]

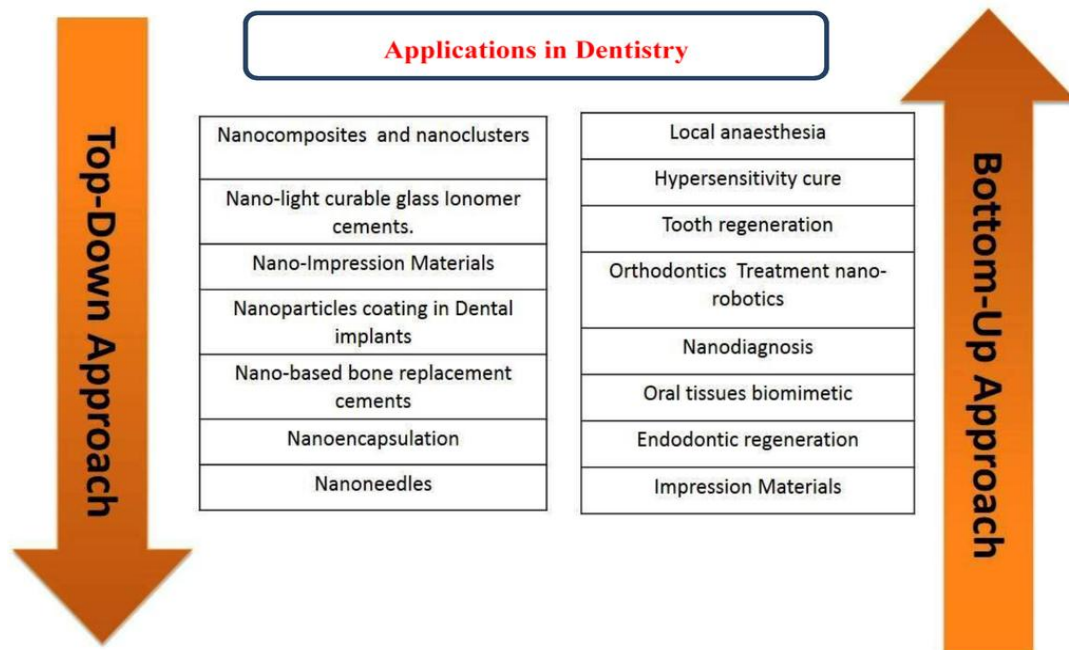


Fig 1.3 Nanoparticle Synthesis Methods

### 1.5.1 TOP-DOWN METHODS

Top-down methods involve breaking down bulk materials into nanoscale particles using mechanical, thermal, or physical techniques. These methods are relatively simple but may produce particles with irregular shapes and defects.

#### Mechanical Milling (Ball Milling)

This method uses high-energy grinding to reduce particle size. Bulk materials are placed in a rotating chamber with hardened steel balls. The repeated impact causes fragmentation into nanoscale particles. Although it is cost-effective and suitable for large-scale production, contamination and poor size control can occur.

#### Lithography Techniques

Lithography methods are widely used in semiconductor fabrication. Techniques such as electron beam lithography create nanoscale patterns on surfaces. These are mainly used in electronic device fabrication rather than powder synthesis.

#### Laser Ablation

In this method, a high-energy laser beam is directed onto a bulk target material. The intense energy vaporizes the material, forming nanoparticles in a surrounding medium. Laser

ablation produces high-purity nanoparticles without chemical reagents. Top-down methods are advantageous for industrial production but often lack precise control over particle size distribution compared to bottom-up methods.

### **1.5.2 BOTTOM-UP METHODS**

Bottom-up approaches are more widely used in research laboratories because they allow better control over particle size, morphology, and composition. These methods involve chemical reactions that assemble nanoparticles from atoms or ions.

#### **Sol-Gel Method**

The sol-gel process involves the transition of a system from a liquid "sol" into a solid "gel." Metal precursors undergo hydrolysis and condensation reactions to form a network structure. This method provides good homogeneity and low-temperature processing.

#### **Co-precipitation Method**

In this method, metal salts are dissolved in solution and precipitated by adjusting pH or adding a precipitating agent. It is simple, cost-effective, and widely used for ferrites and metal oxides. However, controlling particle size requires careful optimization.

#### **Hydrothermal Method**

Hydrothermal synthesis involves chemical reactions in a sealed autoclave at elevated temperature and pressure. It produces highly crystalline nanoparticles with controlled morphology. This method is especially suitable for perovskite and ferrite materials.

#### **Chemical Vapor Deposition (CVD)**

CVD involves vapor-phase chemical reactions that deposit nanoparticles or thin films on substrates. It is widely used in electronic and thin-film industries.

#### **Green Synthesis**

This environmentally friendly method uses plant extracts or biological agents as reducing and stabilizing agents. It avoids toxic chemicals and is gaining importance in biomedical applications.

## **1.6 PEROVSKITES – AN OVERVIEW**

Perovskites are a broad class of materials that possess the general chemical formula  $ABO_3$  and crystallize in a specific structure originally identified in the mineral Perovskite ( $CaTiO_3$ ). The structure was first discovered in the Ural Mountains and later became the prototype for a vast family of compounds known as perovskite materials. Over time, the term

“perovskite” has expanded to include any material that adopts this characteristic crystal structure, regardless of its exact chemical composition. The ideal perovskite structure is cubic, belonging to the space group  $Pm\bar{3}m$ . In this arrangement, the larger A-site cation occupies the corners of the cube, the smaller B-site cation is located at the center, and oxygen anions are positioned at the face centers. The B-site cation is surrounded by six oxygen atoms, forming an octahedral coordination known as  $BO_6$  octahedra. These octahedra are the fundamental building blocks of the perovskite structure and are responsible for many of its physical properties.

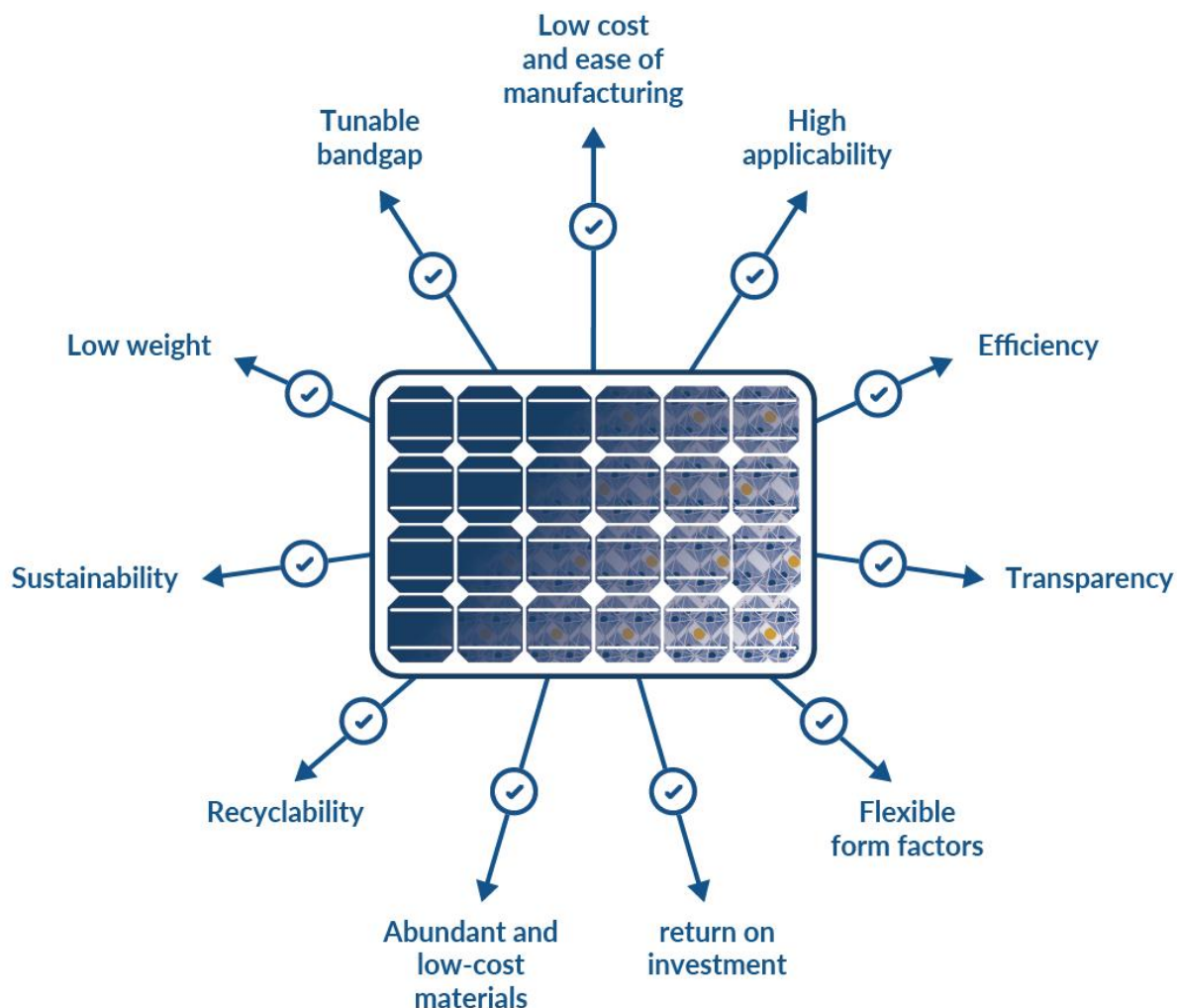
One of the most important characteristics of perovskites is their structural flexibility. A wide range of elements can occupy both the A and B sites, provided they satisfy the tolerance factor condition. The Goldschmidt tolerance factor determines whether a stable perovskite structure can form. Slight deviations from the ideal value result in structural distortions such as orthorhombic, tetragonal, or rhombohedral phases. These distortions significantly influence electrical, magnetic, and optical properties. Perovskites are categorized into several types, including oxide perovskites (such as ferrites and titanates), halide perovskites, and double perovskites. Oxide perovskites, particularly rare-earth orthoferrites, have attracted attention for their magnetic and catalytic properties. Halide perovskites, on the other hand, have revolutionized solar cell research due to their excellent light absorption and charge transport properties. Another important feature of perovskites is their ability to undergo substitution or doping. By replacing A-site or B-site ions with different cations, researchers can tailor the material’s electronic structure, band gap, conductivity, and magnetic behaviour. This tunability makes perovskites highly versatile materials for advanced technological applications. In recent decades, perovskites have gained enormous scientific interest due to their multifunctional nature. They exhibit ferroelectricity, superconductivity, magnetoresistance, and catalytic activity within a single structural family. This combination of properties is rarely observed in other material systems. From an energy perspective, perovskites are considered promising candidates for photocatalysis, water splitting, and solid oxide fuel cells. Their band gap can be engineered to absorb visible light, making them suitable for solar-driven reactions. Furthermore, their thermal and chemical stability enhances their practical applicability in harsh operating conditions. Thus, perovskites represent one of the most important and extensively studied classes of functional materials in modern materials science and condensed matter physics.

## 1.7 PROPERTIES OF PEROVSKITES

Perovskites exhibit a wide range of physical and chemical properties due to their flexible crystal structure and compositional diversity. The interplay between lattice structure, electron configuration, and ionic substitution leads to remarkable multifunctionality. Their properties can be broadly categorized into structural, electrical, optical, magnetic, thermal, and catalytic characteristics. Structurally, perovskites can exist in cubic, tetragonal, orthorhombic, or rhombohedral phases depending on ionic size mismatch and temperature. Structural distortion influences the overlap between electronic orbitals, which directly affects conductivity and magnetic exchange interactions. Even slight tilting of  $\text{BO}_6$  octahedra can dramatically alter physical behaviour. Electrical properties of perovskites vary from insulating to metallic and even superconducting behaviour. Some perovskites exhibit semiconducting characteristics with tunable band gaps, making them suitable for optoelectronic devices. In oxide perovskites, electron hopping between transition metal ions contributes to electrical conductivity. Doping can increase charge carrier concentration and enhance conductivity. Optically, perovskites show strong light absorption across ultraviolet and visible regions. The band gap energy can be modified by adjusting the B-site transition metal or introducing dopants. This property is particularly important for photocatalytic and photovoltaic applications. Their high absorption coefficient and long carrier diffusion length improve efficiency in solar energy conversion.

Magnetic properties in perovskites arise mainly from transition metal ions at the B-site. Exchange interactions between neighbouring ions through oxygen (super exchange mechanism) result in ferromagnetic or antiferromagnetic ordering. Rare-earth orthoferrites, for instance, exhibit interesting spin reorientation transitions and magneto-electric coupling effects. Thermally, perovskites are known for their high stability and resistance to decomposition at elevated temperatures. This makes them suitable for solid oxide fuel cells and catalytic applications. Thermal expansion and phase transitions can occur at specific temperatures, influencing performance in practical devices. Chemically, perovskites possess excellent catalytic activity due to the presence of transition metal ions and oxygen vacancies. Oxygen vacancies enhance ionic mobility and improve redox reactions. These features are particularly beneficial for applications such as water splitting, hydrogen evolution, and oxygen reduction reactions. Another significant property is ferroelectricity in certain perovskites, where spontaneous electric polarization can be reversed by an external electric field. This property is widely used in memory devices, sensors, and actuators. Overall, the multifunctional nature of perovskites stems from the strong coupling between structural, electronic, and

magnetic degrees of freedom. This makes them ideal candidates for next-generation energy and electronic technologies.



**Fig 1.4** Introduction to Perovskites

## 1.8 OBJECTIVE OF THE PRESENT WORK

The present work is focused on the synthesis and investigation of Ag–Sm–Fe–O<sub>3</sub> (silver- and samarium-modified iron oxide with perovskite-type structure) for advanced functional applications. The primary objective is to develop a nanostructured material with improved structural, optical, and catalytic properties suitable for energy-related applications, particularly photocatalysis and water splitting. The first objective of this work is to synthesize Ag/SmFeO<sub>3</sub> nanomaterials using the hydrothermal method. This method is chosen to achieve high crystallinity, controlled particle size, and uniform distribution of constituent elements. Emphasis is placed on optimizing synthesis parameters such as temperature, reaction time, and precursor concentration to obtain phase-pure material. Another important objective is to confirm the formation of the perovskite structure and analyze the crystal phase of the

synthesized material. Structural characterization aims to verify the incorporation of silver and samarium ions into the iron oxide lattice and to study the effect of these modifications on lattice distortion and crystallinity. The present work also aims to investigate the morphological features of the synthesized Ag/SmFeO<sub>3</sub> nanoparticles. Particle size, shape, and surface uniformity play a crucial role in determining catalytic activity. Therefore, understanding morphology is essential for correlating structural features with functional performance. A key objective of this study is to analyze the optical properties of Ag/SmFeO<sub>3</sub> nanomaterials. Special attention is given to band gap evaluation and light absorption behavior. Modification with silver and samarium is expected to enhance visible-light absorption, which is essential for efficient photocatalytic applications. Another objective is to study the role of silver incorporation in improving charge carrier separation and reducing electron–hole recombination. Silver is known to act as an electron trap, thereby enhancing photocatalytic efficiency. The interaction between silver and the perovskite lattice is examined to understand its influence on material performance.

The work further aims to evaluate the photocatalytic potential of Ag/SmFeO<sub>3</sub> for water splitting applications. The focus is on assessing hydrogen generation efficiency under light irradiation and understanding the mechanism of photocatalytic activity based on structural and optical properties. Additionally, the study seeks to establish a structure–property–performance relationship for Ag/SmFeO<sub>3</sub> nanomaterials. By correlating synthesis conditions, crystal structure, morphology, and optical behavior, the research aims to identify key factors responsible for enhanced catalytic activity. Finally, the overall objective of the present work is to contribute to the development of efficient, stable, and low-cost perovskite-based nanomaterials for sustainable energy applications. The findings of this study are expected to provide valuable insights into the design of modified perovskite oxides for future photocatalytic and energy conversion technologies.

## CHAPTER II

### REVIEW OF LITERATURE

Perovskite-type oxides with the general chemical formula  $ABO_3$  have been extensively investigated due to their exceptional structural flexibility and wide range of physical properties. The perovskite structure, first identified in the mineral calcium titanate, consists of a three-dimensional network of corner-sharing  $BO_6$  octahedra, where the smaller B-site cation is coordinated by six oxygen atoms, while the larger A-site cation occupies the interstitial spaces between these octahedra. The ideal structure is cubic; however, most practical perovskites exhibit distorted structures such as orthorhombic or tetragonal symmetry due to differences in ionic radii. The stability and distortion of the perovskite lattice are commonly explained using the Goldschmidt tolerance factor, which relates the ionic radii of the constituent ions. When the tolerance factor deviates from unity, structural distortion occurs through octahedral tilting, which modifies bond angles and lattice parameters. These structural variations strongly influence crystallographic properties and are commonly detected through X-ray diffraction studies [1,2,3].

Rare-earth orthoferrites with the general formula  $RFeO_3$  (where R represents a rare-earth element) form an important subgroup of perovskite oxides. These materials typically crystallize in an orthorhombic structure belonging to the  $Pbnm$  space group. In these compounds,  $Fe^{3+}$  ions occupy the B-site positions and form  $FeO_6$  octahedra, while rare-earth ions such as  $Sm^{3+}$  occupy the A-site. The difference in ionic size between rare-earth elements significantly influences lattice distortion and structural symmetry.  $SmFeO_3$  has received particular attention because the relatively smaller ionic radius of  $Sm^{3+}$  produces noticeable distortion in the perovskite framework. This distortion modifies Fe–O–Fe bond angles and leads to detectable shifts in diffraction peaks during structural analysis. Previous research has demonstrated that such distortions contribute to microstrain within the lattice, which manifests as peak broadening in XRD patterns [4,5,6]

Extensive research has been carried out on the synthesis of rare-earth orthoferrites using various preparation methods, including conventional solid-state reaction, sol–gel process, combustion synthesis, co-precipitation, and hydrothermal techniques. Among these methods, hydrothermal synthesis has gained considerable attention due to its ability to produce highly

crystalline nanoparticles at relatively lower temperatures compared to solid-state methods. The hydrothermal technique involves chemical reactions in aqueous solution under elevated temperature and pressure inside a sealed autoclave, which promotes enhanced solubility of precursors and controlled nucleation of particles. Researchers have reported that reaction temperature, pH value, precursor concentration, and reaction time significantly influence phase formation, crystallinity, and particle morphology. The controlled environment provided by hydrothermal processing enables formation of uniform nanocrystalline materials with reduced impurity phases [7,8].

At the nanoscale, materials exhibit distinct structural characteristics compared to their bulk counterparts. Reduction in crystallite size leads to increased surface-to-volume ratio and enhanced grain boundary area. In X-ray diffraction analysis, nanoscale particles typically show peak broadening due to reduced crystallite size and internal lattice strain. The Debye–Scherrer equation is widely used to estimate average crystallite size from the full width at half maximum of diffraction peaks. Several investigations have confirmed that nanocrystalline  $\text{SmFeO}_3$  exhibits broadened diffraction peaks compared to bulk samples, indicating successful nanoscale formation. In addition to size effects, lattice strain arising from ionic substitution or structural distortion also contributes to peak broadening. Therefore, detailed XRD analysis is essential for confirming phase purity, crystal symmetry, and nanoscale crystallite dimensions. [9,10,11]

Ionic substitution or doping is frequently employed to modify the structural characteristics of perovskite oxides. Substitution at the A-site or B-site can alter lattice parameters, induce strain, and modify bond geometry. The incorporation of metal ions with different ionic radii leads to lattice distortion and changes in interatomic spacing. Silver incorporation into perovskite systems has been reported to influence structural parameters by introducing strain within the lattice. Due to the difference in ionic radius and valence state between silver ions and rare-earth ions, substitution may lead to charge imbalance and possible formation of oxygen vacancies. Such structural modifications are often reflected in subtle shifts of diffraction peaks and changes in peak intensity. Researchers have emphasized that these changes are strong indicators of successful dopant incorporation within the perovskite framework. [12,13,14]

Morphological analysis plays a crucial role in understanding the microstructural features of synthesized nanoparticles. Scanning Electron Microscopy (SEM) is widely used to investigate particle shape, grain distribution, and surface texture. Literature reports indicate that hydrothermally synthesized rare-earth ferrites commonly exhibit irregular or nearly spherical morphology with moderate agglomeration. Agglomeration is a common phenomenon in nanomaterials due to high surface energy and strong interparticle forces such as van der Waals attraction and magnetic interactions. However, controlled synthesis conditions can minimize excessive clustering and promote relatively uniform grain formation. The presence of surfactants or chelating agents during synthesis significantly influences particle growth and morphology. [15,16,17]

Chelating agents such as Urea are commonly used to ensure homogeneous mixing of metal ions at the molecular level. These agents form stable metal–ligand complexes, preventing premature precipitation and ensuring uniform distribution of cations. During hydrothermal treatment, decomposition of these complexes leads to controlled nucleation and growth of nanoparticles. Previous studies have reported that Urea -assisted synthesis results in improved phase purity, enhanced crystallinity, and reduced agglomeration. The combination of controlled nucleation and restricted grain growth contributes to formation of fine nanoparticles with uniform distribution. [18,19,20]

Correlation between structural and morphological analysis is essential for a comprehensive understanding of nanomaterials. Crystallite size estimated from XRD analysis often corresponds closely with particle size observed in SEM images. Slight discrepancies between crystallite size and grain size may arise due to particle agglomeration or presence of multiple crystallites within a single grain. Several researchers have emphasized that combining XRD and SEM techniques provides reliable confirmation of nanoscale particle formation and phase purity. [21,22]

Although numerous studies have been conducted on rare-earth orthoferrites, comparatively fewer investigations focus on silver-modified  $\text{SmFeO}_3$  synthesized through hydrothermal methods with primary emphasis on structural and morphological characterization. Many earlier works concentrated on magnetic or optical properties, while detailed evaluation of lattice distortion and microstructural evolution due to silver incorporation remains limited. Therefore, systematic structural and morphological

investigation of Ag-incorporated  $\text{SmFeO}_3$  nanoparticles is necessary to understand the influence of dopant incorporation on crystal symmetry and grain formation. The present study builds upon existing literature by synthesizing  $\text{AgSmFeO}_3$  nanoparticles using hydrothermal technique and analyzing their phase formation, crystallinity, lattice strain, and surface morphology using XRD and SEM techniques, thereby contributing to the ongoing research on nanostructured perovskite materials. [23,24,25]

## REFERENCES

1. Goldschmidt, V.M., 1926. "Die Gesetze der Krystallochemie," *Naturwissenschaften*. Introduced tolerance factor concept explaining structural stability of perovskites.
2. Debye, P., Scherrer, P., 1916. "Interference of X-rays in Crystals," *Physikalische Zeitschrift*. Developed Scherrer equation for calculating crystallite size from XRD peak broadening.
3. Goodenough, J.B., 1963. *Magnetism and the Chemical Bond*. Explained superexchange interaction in transition metal oxides including orthoferrites.
4. Shannon, R.D., 1976. "Revised Effective Ionic Radii," *Acta Crystallographica A*. Provided ionic radius data widely used in perovskite structural analysis.
5. Geller, S., 1956. "Crystal Structure of Rare Earth Orthoferrites," *Journal of Chemical Physics*. Reported structural characteristics of  $\text{RFeO}_3$  compounds.
6. Peña, M.A., Fierro, J.L.G., 2001. "Chemical Structures and Performance of Perovskites," *Chemical Reviews*. Reviewed structure–property relationships in perovskite oxides.
7. Cullity, B.D., Stock, S.R., 2001. *Elements of X-Ray Diffraction*. Standard reference for XRD analysis and crystallite size determination.
8. Klug, H.P., Alexander, L.E., 1974. *X-Ray Diffraction Procedures*. Described detailed procedures for phase identification and line broadening analysis.
9. Byrappa, K., Yoshimura, M., 2001. *Handbook of Hydrothermal Technology*. Explained principles and applications of hydrothermal synthesis.
10. Eerenstein, W., Mathur, N.D., Scott, J.F., 2006. "Multiferroic Materials," *Nature*. Discussed multifunctional properties of perovskite oxides.
11. Sugimoto, T., 2000. "Monodispersed Particles," *Journal of Colloid Interface Science*. Reported methods to control particle size and morphology.

12. Wang, Z.L., 1998. "Characterization of Nanophase Materials," *Advanced Materials*.  
Discussed structural and morphological analysis of nanomaterials.
13. Patil, K.C., et al., 2002. "Combustion Synthesis of Oxides," *Current Opinion in Solid State Materials Science*.  
Compared synthesis methods for oxide nanoparticles.
14. Battle, P.D., 1989. "Structural Chemistry of Perovskites," *Journal of Solid State Chemistry*.  
Studied structural distortions in perovskite systems.
15. Williamson, G.K., Hall, W.H., 1953. "X-ray Line Broadening," *Acta Metallurgica*.  
Introduced method to separate strain and size broadening in XRD analysis.
16. Rao, C.N.R., Gopalakrishnan, J., 1997. *New Directions in Solid State Chemistry*.  
Described structural chemistry of transition metal oxides.
17. Chen, X., Mao, S.S., 2007. "Nanomaterials for Applications," *Chemical Reviews*.  
Reviewed synthesis and characterization of nanomaterials.
18. Li, W., et al., 2010. "Hydrothermal Synthesis of Ferrites," *Materials Letters*.  
Reported hydrothermal preparation of rare-earth ferrite nanoparticles.
19. Zhang, L., et al., 2012. "Nanostructured Perovskite Oxides," *Journal of Alloys and Compounds*.  
Studied structural and morphological features of doped perovskites.
20. Singh, J., et al., 2015. "Structural Study of SmFeO<sub>3</sub> Nanoparticles," *Ceramics International*.  
Investigated XRD and SEM characteristics of SmFeO<sub>3</sub> nanoparticles.
21. Sun, S., Murray, C.B., 1999. "Synthesis of Monodisperse Nanoparticles," *Journal of Applied Physics*.  
Discussed control of particle growth during synthesis.
22. Rodriguez-Carvajal, J., 1993. "Magnetic Structure Determination," *Physica B*.  
Reported structural refinement methods for magnetic oxides.
23. Blasse, G., 1965. "Magnetic Properties of Orthoferrites," *Philips Research Reports*.  
Studied magnetic ordering in rare-earth ferrites.
24. Tauc, J., 1968. "Optical Properties of Semiconductors," *Materials Research Bulletin*.  
Introduced Tauc relation for band gap estimation.
25. Feynman, R.P., 1959. "There's Plenty of Room at the Bottom."  
Inspired nanoscale research and development in materials science.

## **CHAPTER III**

### **METHODOLOGY**

#### **3.1 MATERIALS AND REAGENTS**

For the synthesis of Ag/SmFeO<sub>3</sub> nanoparticles, high-purity analytical grade precursor chemicals were used in order to ensure phase purity and avoid contamination during the synthesis process. Silver precursor, samarium precursor, and iron precursor were selected as metal sources for A-site and B-site cations in the perovskite structure. Urea was used as a surfactant and complexing agent. Deionized water was employed as the solvent because it minimizes unwanted ionic interference and ensures uniform dissolution of precursor salts. The selection of these specific precursors is important because the hydrothermal synthesis method requires complete solubility of metal ions to promote homogeneous nucleation. Any impurity or incomplete dissolution may result in secondary phases or incomplete perovskite formation. Therefore, all chemicals were measured accurately using a digital analytical balance with high precision.

#### **3.2 HYDROTHERMAL SYNTHESIS OF Ag/SmFeO<sub>3</sub> NANOPARTICLES**

The hydrothermal method was chosen for synthesizing Ag/SmFeO<sub>3</sub> nanoparticles due to its ability to provide controlled nucleation, uniform particle growth, improved crystallinity, and lower synthesis temperature compared to conventional solid-state methods. Hydrothermal synthesis involves chemical reactions in aqueous solution under high temperature and pressure conditions inside a sealed autoclave system. This method enhances solubility of precursors and accelerates crystallization kinetics, leading to well-defined nanostructures.

#### **3.3 PREPARATION OF SILVER PRECURSOR SOLUTION**

A calculated amount of 1.6 g of silver precursor corresponding to 0.5 molar concentration was dissolved in 20 mL of deionized water. The dissolution process was carried out under continuous magnetic stirring to ensure complete ionization of silver species into Ag<sup>+</sup> ions. Uniform stirring prevents localized precipitation and ensures homogeneous ionic distribution throughout the solution. Silver ions play a crucial role in modifying structural, optical, and electronic properties of the final Ag/SmFeO<sub>3</sub> compound. Since Ag has a different valence state compared to Sm<sup>3+</sup> and Fe<sup>3+</sup>, its incorporation into the lattice is expected to induce charge imbalance, which may be compensated either by oxygen vacancy formation or partial

oxidation of  $\text{Fe}^{3+}$  to  $\text{Fe}^{4+}$ . Therefore, proper dissolution and homogeneous mixing of silver precursor is essential to achieve uniform substitution within the perovskite lattice.

### **3.4 PREPARATION OF SAMARIUM PRECURSOR SOLUTION**

Separately, 1.12 g of samarium precursor corresponding to 0.25 molar concentration was dissolved in 10 mL of deionized water under continuous magnetic stirring.  $\text{Sm}^{3+}$  ions occupy the A-site of the perovskite structure and strongly influence lattice distortion due to their ionic size. The dissolution of samarium salt ensures uniform distribution of  $\text{Sm}^{3+}$  ions in the reaction medium. Since rare-earth ions have larger ionic radii compared to transition metal ions, their presence controls the tolerance factor and structural symmetry of the final product. Proper dissolution prevents formation of samarium hydroxide precipitates, which could otherwise affect phase purity.

### **3.5 PREPARATION OF IRON PRECURSOR SOLUTION**

Similarly, 1 g of iron precursor corresponding to 0.25 molar concentration was dissolved in 10 mL of deionized water with constant magnetic stirring.  $\text{Fe}^{3+}$  ions form the backbone of the perovskite lattice by occupying the B-site and forming  $\text{FeO}_6$  octahedral units. Iron ions are responsible for magnetic ordering and electronic conduction in orthoferrite compounds. The superexchange interaction between adjacent  $\text{Fe}^{3+}$  ions through oxygen anions determines the magnetic nature of the material. Therefore, achieving homogeneous dispersion of  $\text{Fe}^{3+}$  ions in the precursor solution is essential for uniform Fe–O–Fe network formation in  $\text{Ag}/\text{SmFeO}_3$ .

### **3.6 ADDITION OF SURFACTANT – ROLE OF UREA**

A measured quantity of 1.9 g of Urea corresponding to 0.25 molar concentration was dissolved in 40 mL of deionized water. Urea acts as a chelating agent, stabilizer, and particle growth controller. The carboxyl ( $-\text{COOH}$ ) and hydroxyl ( $-\text{OH}$ ) functional groups present in Urea form coordination bonds with metal ions such as  $\text{Ag}^+$ ,  $\text{Sm}^{3+}$ , and  $\text{Fe}^{3+}$ . This chelation process prevents premature precipitation of metal hydroxides and ensures molecular-level mixing of all cations. Additionally, Urea reduces agglomeration by forming a protective organic layer around nucleating particles. During hydrothermal treatment, decomposition of Urea releases gases and facilitates controlled nucleation, leading to fine and uniform nanoparticles. Thus, the surfactant plays a critical role in controlling particle size, morphology, and structural homogeneity of  $\text{Ag}/\text{SmFeO}_3$ .

### 3.7 MIXING AND MAGNETIC STIRRING PROCESS

All prepared precursor solutions were slowly mixed together under continuous magnetic stirring. The final mixed solution had a total volume of approximately 80 mL. The stirring process was maintained for several hours to achieve complete homogeneity. During mixing, metal–citrate complexes are formed, ensuring even distribution of  $\text{Ag}^+$ ,  $\text{Sm}^{3+}$ , and  $\text{Fe}^{3+}$  ions. Proper mixing avoids phase segregation and ensures stoichiometric balance. The pH of the solution becomes stabilized during this process, which is important because pH influences nucleation rate and crystal growth mechanism. The formation of a homogeneous precursor solution is a critical step in hydrothermal synthesis, as it determines the phase purity and uniformity of the final  $\text{AgSmFeO}_3$  nanoparticles.

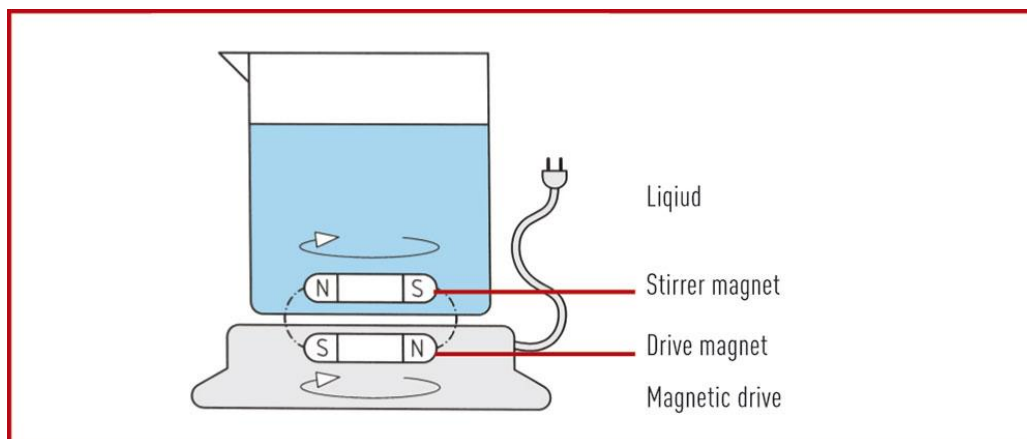


Fig 3.1 Mixing and Magnetic Stirring Process

### 3.8 HYDROTHERMAL TREATMENT IN TEFLON-LINED AUTOCLAVE

The homogeneous precursor solution was transferred into a Teflon-lined stainless steel autoclave with a capacity of 80 mL. The Teflon liner prevents corrosion and contamination under high-temperature and high-pressure conditions. The sealed autoclave was placed in a hot air oven maintained at  $180^\circ\text{C}$  for 12 hours. Under hydrothermal conditions, water acts as both solvent and reaction medium. At elevated temperature, autogenous pressure develops inside the sealed vessel, enhancing solubility and diffusion of ions. The high-temperature environment promotes nucleation and crystal growth of  $\text{Ag}/\text{SmFeO}_3$  nanoparticles. Metal hydroxides formed initially undergo dehydration and rearrangement to form the orthorhombic perovskite structure. Controlled temperature and reaction time ensure proper crystallinity and phase formation.

### **3.9 COOLING, WASHING AND CENTRIFUGATION**

After completion of the reaction, the autoclave was allowed to cool naturally to room temperature. Gradual cooling prevents structural stress and microcrack formation. The obtained precipitate was collected and washed multiple times with deionized water to remove unreacted ions and residual Urea. The suspension was subjected to centrifugation at high speed, allowing solid nanoparticles to settle at the bottom while the supernatant liquid was discarded. Centrifugation ensures purification of nanoparticles and removal of soluble impurities. This step is crucial to obtain phase-pure Ag/SmFeO<sub>3</sub> powder suitable for characterization studies.

### **3.10 DRYING AND POWDER COLLECTION**

The purified precipitate was dried in a hot air oven at moderate temperature to remove residual moisture. The dried product was gently ground using an agate mortar and pestle to obtain fine nanopowder. The final product obtained was Ag/SmFeO<sub>3</sub> nanopowder with improved crystallinity, controlled morphology, and reduced agglomeration due to hydrothermal processing and surfactant assistance.

## CHAPTER IV

### CHARACTERIZATION TECHNIQUES

#### 4.1 X-Ray Diffraction (XRD)

X-Ray Diffraction (XRD) is one of the most important and widely used analytical techniques for studying the structural properties of crystalline materials, especially in the field of nanomaterials and solid-state physics. XRD is primarily used to determine the crystal structure, phase identification, lattice parameters, crystallite size, and degree of crystallinity of a material. The basic principle of XRD is based on the diffraction of X-rays by the periodic arrangement of atoms in a crystal lattice. When a beam of monochromatic X-rays with a specific wavelength is directed toward a crystalline sample, the atoms in the crystal act as scattering centers and cause the incident X-rays to diffract in different directions. Constructive interference of these diffracted beams occurs only when the path difference between reflected waves satisfies Bragg's Law, which is given by  $n\lambda = 2d \sin\theta$ , where  $n$  is the order of diffraction,  $\lambda$  is the wavelength of the incident X-ray,  $d$  is the interplanar spacing between crystal planes, and  $\theta$  is the diffraction angle. By measuring the angles and intensities of the diffracted beams, the crystal structure and other structural characteristics of the material can be determined. In a typical XRD instrument, the major components include an X-ray source (usually Cu-K $\alpha$  radiation), a sample holder, a goniometer to control the angles of incidence and detection, and a detector that records the intensity of diffracted X-rays. During the analysis, the sample is exposed to X-rays while the detector moves around it, collecting the diffraction pattern over a range of angles (usually represented as  $2\theta$ ). The resulting diffraction pattern consists of a series of peaks at specific angles corresponding to different crystal planes in the material. Each crystalline material produces a unique diffraction pattern, which acts as a "fingerprint" for identifying the material and confirming its phase purity. In nanomaterials research, XRD is particularly important because it allows researchers to estimate the average crystallite size using the Scherrer equation, which relates peak broadening in the diffraction pattern to the size of the crystalline domains. Additionally, XRD can provide information about strain, lattice distortion, and structural defects present in the crystal lattice. The intensity and sharpness of the peaks also indicate the degree of crystallinity of the synthesized material. In the study of perovskite nanomaterials such as Ag-doped SmFeO<sub>3</sub>, XRD plays a crucial role in confirming the formation of the perovskite phase, verifying successful doping, and ensuring that no secondary impurity phases are present. Thus, XRD serves as a powerful, non-destructive

technique for structural characterization and is widely used in materials science, chemistry, physics, and nanotechnology research.



Fig 4.1 X-Ray Diffraction (XRD) instrument

#### **4.2 Scanning Electron Microscopy (SEM)**

Scanning Electron Microscopy (SEM) is a powerful imaging technique widely used to study the surface morphology, microstructure, and particle size of materials at the nanoscale and microscale. Unlike optical microscopes, which use visible light for imaging, SEM uses a focused beam of high-energy electrons to scan the surface of a specimen and produce highly magnified images with excellent resolution and depth of field. The working principle of SEM is based on the interaction between the incident electron beam and the atoms present on the surface of the sample. When the electron beam strikes the sample surface, various signals are generated due to electron-matter interactions, including secondary electrons, backscattered electrons, and characteristic X-rays. Among these signals, secondary electrons are most commonly used for imaging because they provide detailed information about the surface topography and morphology of the material. In a typical SEM instrument, the main components include an electron gun (which generates the electron beam), electromagnetic lenses to focus the beam, scanning coils to move the beam across the sample surface, a sample chamber maintained under high vacuum, and detectors that collect the emitted signals. During the analysis, the electron beam is raster-scanned across the surface of the specimen, and the emitted electrons are detected and converted into electrical signals, which are then processed to form a high-resolution image on a computer screen. SEM images can reveal important information

such as particle size, shape, surface texture, grain boundaries, porosity, and the degree of agglomeration of nanoparticles. In nanomaterials research, SEM is especially valuable for understanding how synthesis conditions affect the morphology of the material. For example, parameters such as temperature, reaction time, and precursor concentration can influence the growth and distribution of nanoparticles, which can be clearly observed using SEM analysis. Additionally, SEM can be coupled with Energy Dispersive X-ray Spectroscopy (EDS or EDX) to perform elemental analysis and confirm the presence of specific elements within the sample. In the characterization of perovskite nanomaterials such as Ag-doped  $\text{SmFeO}_3$  nanoparticles, SEM helps to visualize the surface structure and morphology, providing insights into the size distribution and agglomeration behavior of the synthesized particles. The images obtained from SEM typically show nanoscale grains or clusters, which may appear slightly agglomerated due to high surface energy of nanoparticles. By analyzing these images, researchers can better understand the structural features and growth mechanisms of the synthesized material. Therefore, SEM is considered an essential characterization technique in nanotechnology and materials science, as it provides detailed morphological information that complements structural analysis obtained from techniques such as X-ray diffraction.



Fig 4.2 Scanning Electron Microscopy (SEM)

## CHAPTER V

### RESULTS AND DISCUSSION

#### 5.1 STRUCTURAL ANALYSIS

The structural properties of Ag/SmFeO<sub>3</sub> samples were studied using X-ray diffractometer. The peaks observed confirms the high crystallinity of the samples produced and the XRD analysis of the sample consists of many diffraction peaks of Ag/SmFeO<sub>3</sub>. The crystalline size of the CeFeO<sub>3</sub> samples were calculated using Debye Scherer's formula as

$$D = \frac{k\lambda}{\beta \cos \theta}$$

Where,

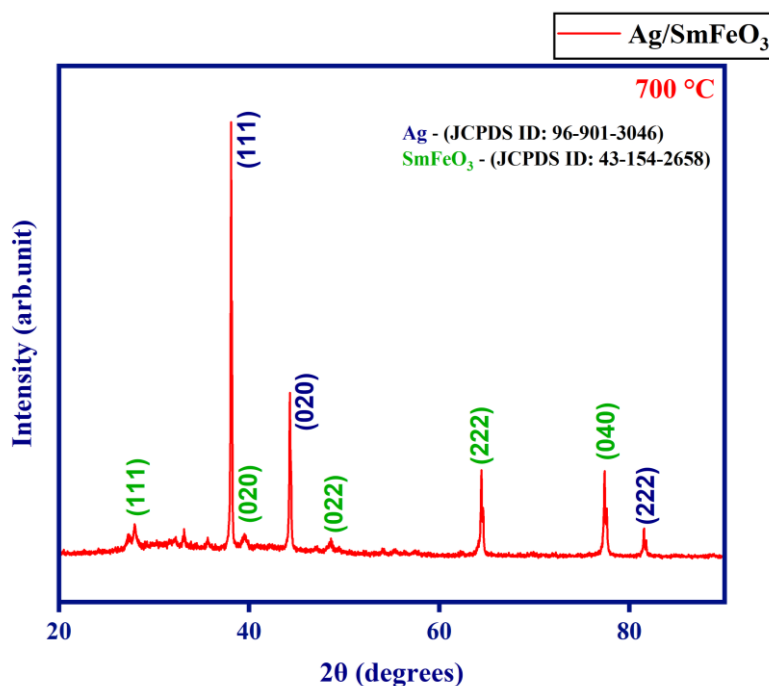
**D** - Size of the crystal in the direction perpendicular to the reflection plane (nm)

**k** - Proportionality constant = 0.94

**λ** - Wavelength of X-ray = 0.15406 (nm) or 1.5406 Å

**β** - The full width at Half Maximum of the peak (radians)

**θ** - Diffraction angle (radians)

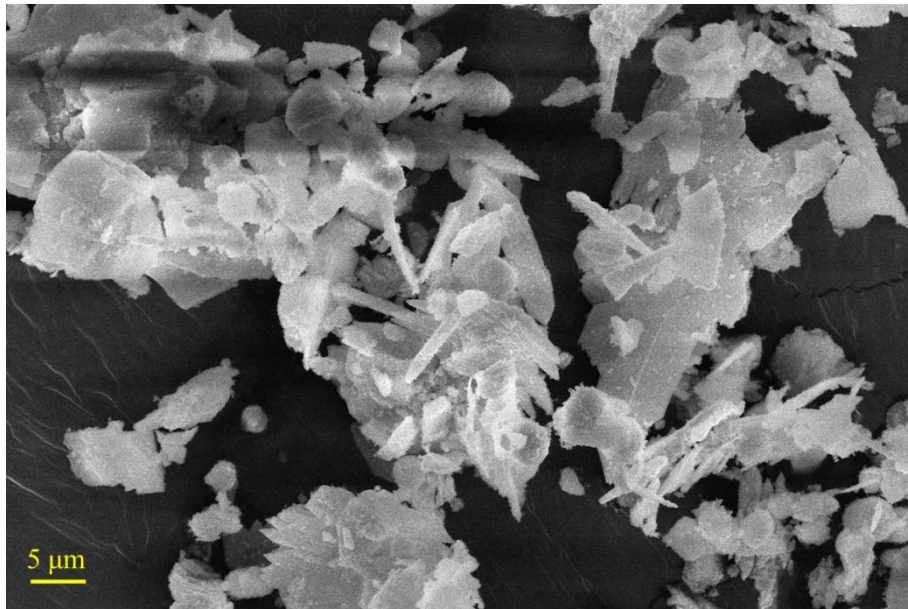


**Fig 5.1 (a)** X-ray diffraction Pattern of Ag/SmFeO<sub>3</sub> nanostructure

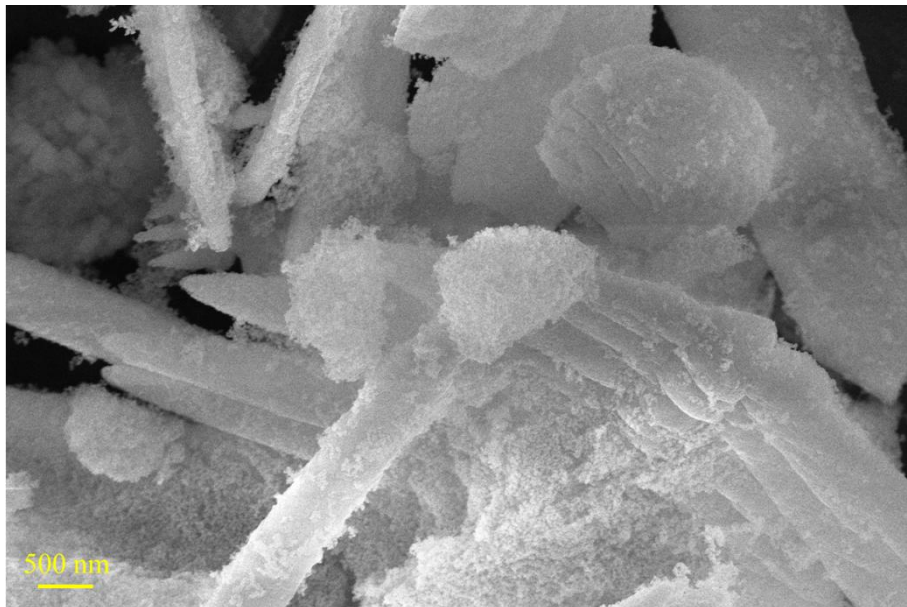
Crystal structure and phase purity of the prepared samples were determined using powder XRD analysis. The XRD patterns of Ag/SmFeO<sub>3</sub> nanostructures (spheres and rectangular) shown in Figure 5.1, revealed the Ag and SmFeO<sub>3</sub> polycrystalline nature and exhibits typical Cubic and perovskite orthorhombic structure respectively (Ag JCPDS File No: 96-901-3046) and (SmFeO<sub>3</sub> JCPDS File No:43-154-2658) for the diffraction peaks at 2 $\theta$  values of 28.55°, 38.09°, 39.49°, 44.35°, 49.107°, 66.436°, 78.769°, and 83.46° assigned to (111), (1111), (020), (020), (022), (222), (040), and (222) planes, respectively. No other and secondary impurity peaks corresponding to Fe-containing or Sm-containing phases were detected in the XRD patterns indicating that the Ag/SmFeO<sub>3</sub> samples of high purity could be obtained under preparation conditions. The strong and sharp diffraction peaks suggest that Ag/SmFeO<sub>3</sub> nanostructure samples are highly crystalline. The average crystallite size of the sphere using the Debye Scherrer formula was calculated and it was found to be ~ 54 nm. The calculated crystallite size was found to be in the nanometer range, confirming successful nanoscale synthesis. Reduction in crystallite size is mainly due to controlled nucleation under hydrothermal conditions and the presence of Urea acting as a chelating agent. The small crystallite size increases the surface-to-volume ratio, which is beneficial for catalytic and surface-related applications.

## 5.2 SURFACE MORPHOLOGY ANALYSIS

Surface morphology of shape-selective synthesis of Ag/SmFeO<sub>3</sub> nanostructures achieved by surfactant like urea shown in Figure 5.2. rectangular nanostructures were obtained when Urea was used as the surfactant as shown in Figure 5.2 (a). It reveals that the individual rectangular has a diameter of ~500 nm as exposed in Figure 5.2 (b). These rectangular-shaped arise from the single-center, thereby attaining the formation of a structure via self-assembly process using the hydrothermal process. We can be observed that an individual rectangular structures is an assembly of many nanoparticles, as presented in Figure 5.2 (b).



**Fig 5.2 (a)** Surface morphology of low magnification of rectangular structures like Ag/SmFeO<sub>3</sub> nanostructure (Urea as Surfactant)



**Fig 5.2 (b)** Surface morphology of high magnification of rectangular structures like Ag/SmFeO<sub>3</sub> nanostructure (Urea as Surfactant)

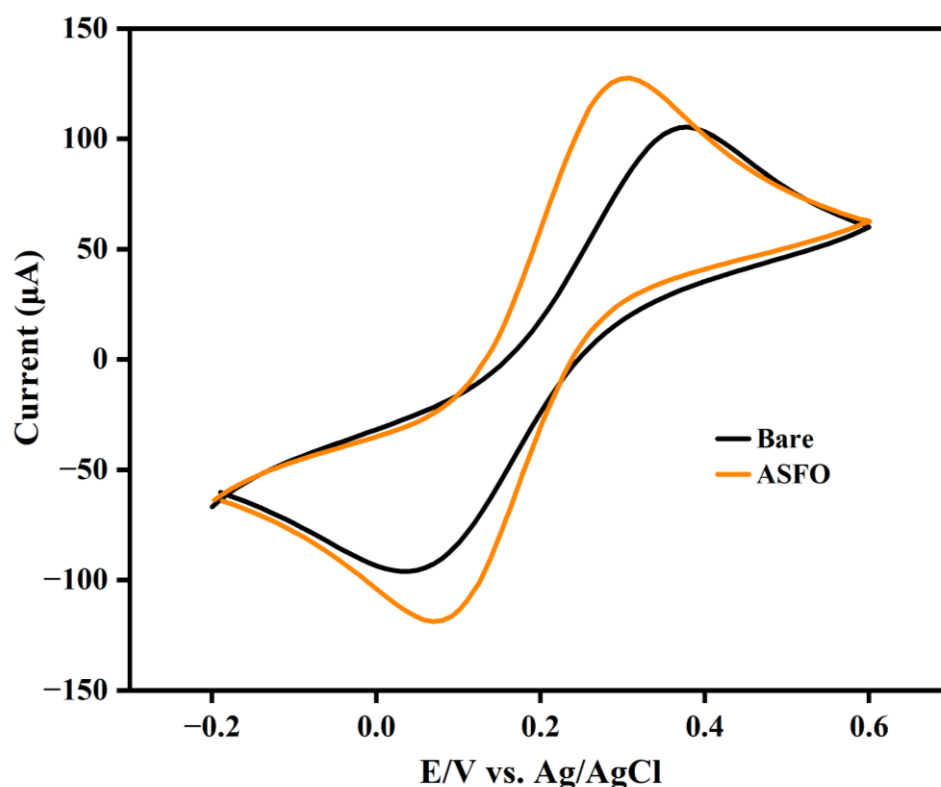
The formation mechanism can be controlled by the intrinsic and extrinsic factors, including the diffusion of the reaction, crystal orientation, surface energy, and the nature of the surfactant in the solution reaction system. urea acts the role of surfactant, OH<sup>-</sup> ions absorb on the surface of Ag/SmFeO<sub>3</sub> formed in the direction of the reaction process and continues until spindles aggregate into rectangular structures via nucleation, aggregation, and recrystallization process were obtained. During the reaction process, the structure direct agent was absorbed by

the growth rate of crystal faces, which thereby controls the size and shape through an Ostwald ripening process. Therefore, Ag/SmFeO<sub>3</sub> nanospheres as the high crystallinity, nanocomposite domain size, and uniform shape should help maintain the highly open porous structure upon deployment of this material, which should favour the of catalytic applications and further characterization.

### 5.3 ELECTROCHEMICAL CHARACTERIZATION OF Ag/SmFeO<sub>3</sub> NANOSTRUCTURE

#### 5.3.1. CYCLIC VOLTAMMETRIC (CV) OF Ag/SmFeO<sub>3</sub> NANOSTRUCTURE

The cyclic voltammetric (CV) responses of the bare glassy carbon electrode (GCE) and GCEs modified with Ag/SmFeO<sub>3</sub> nanostructures were recorded in a 10 mM [Fe(CN)<sub>6</sub>]<sup>3-/4-</sup> solution with 1.0 M KCl as the supporting electrolyte. The CV curves, as shown in Figure 5.3, reveal a significant enhancement in current response at the Ag/SmFeO<sub>3</sub>-modified GCE compared to the bare GCE. This improvement is attributed to the larger electrochemically active surface area (ECSA) and the efficient electron transfer pathways provided by the surfactant-assisted synthesis.



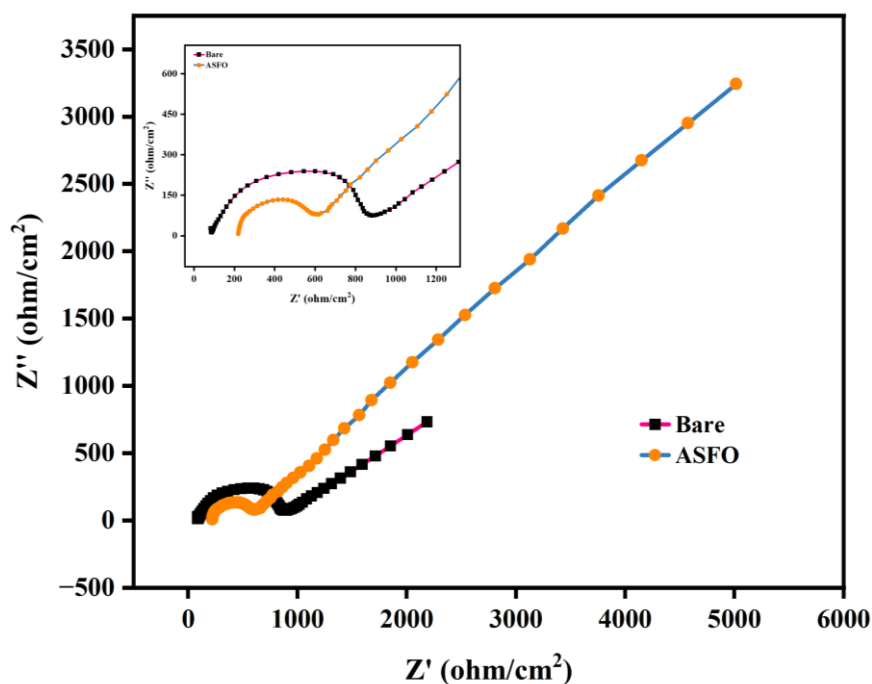
**Fig 5.3** Cyclic voltammetric (CV) of the bare glassy carbon electrode and Ag/SmFeO<sub>3</sub>-modified electrodes.

The increase in peak current at ASFO @GCE confirms that this electrode modification strategy optimizes electrochemical performance. The exceptional efficiency of ASFO is a result of the synergistic effect between silver (Ag) and samarium iron oxide ( $\text{SmFeO}_3$ ). While Ag enhances conductivity, the perovskite oxide framework of  $\text{SmFeO}_3$  provides excellent catalytic activity, leading to faster electron transfer kinetics and increased sensitivity. This combination of superior electrical conductivity, high surface area, and enhanced charge transfer efficiency makes Ag/ $\text{SmFeO}_3$  an ideal electrode material for developing next-generation electrochemical sensors capable of detecting trace-level analytes with high sensitivity and selectivity.

### **5.3.2. ELECTROCHEMICAL IMPEDANCE SPECTROSCOPY (EIS) OF Ag/ $\text{SmFeO}_3$ NANOSTRUCTURE**

Electrochemical impedance spectroscopy (EIS) is a widely utilized technique to analyze the charge transfer characteristics and electron transport dynamics of modified electrodes. In a typical Nyquist plot, two distinct features are observed: a semicircular region at higher frequencies corresponding to the charge transfer resistance ( $R_{ct}$ ) and a linear segment at lower frequencies representing the diffusion-controlled process. The diameter of the semicircle is a crucial indicator of the surface electron transfer resistance, which directly affects electrochemical performance.

To evaluate the electrochemical behaviour of the synthesized Ag/ $\text{SmFeO}_3$  nanostructures, EIS measurements were carried out using a 5 mM  $[\text{Fe}(\text{CN})_6]^{3-/4-}$  solution in 0.1 M KCl as the electrolyte. Various electrode configurations were examined, including a bare glassy carbon electrode (black) and GCEs modified with Ag/ $\text{SmFeO}_3$  nanostructures. The EIS spectra, as presented in Figure 5.4, illustrate notable differences in electron transfer efficiency across these modified electrodes. The bare GCE exhibited a significantly large semicircle, indicating high charge transfer resistance and poor electron transport efficiency. Conversely, the most pronounced enhancement in charge transfer was observed in the Ag/ $\text{SmFeO}_3$ -modified electrodes, as evidenced by their smaller semicircle diameters, which correlate with superior electron transport properties.



**Fig 5.4** EIS curves of the bare glassy carbon electrode and Ag/SmFeO<sub>3</sub>-modified electrodes.

The R<sub>ct</sub> values for the different electrodes were calculated as follows:

- Bare GCE: 824.6  $\Omega$
- ASFO @GCE: 366.2  $\Omega$

Similarly, the R<sub>s</sub> values were determined as:

- Bare GCE: 87.8  $\Omega$
- ASFO @GCE: 228.3  $\Omega$

Among these configurations, the Ag/SmFeO<sub>3</sub>-modified electrode exhibited the lowest R<sub>ct</sub>, confirming its superior electron transfer capability. The enhanced conductivity of Ag/SmFeO<sub>3</sub> is attributed to its highly porous morphology, which facilitates efficient electron transport pathways and increases the electrode's electrochemically active surface area. These findings highlight the potential of Ag/SmFeO<sub>3</sub> nanostructures as a promising electrode material for advanced electrochemical applications.

## CHAPTER VI

### SUMMARY AND CONCLUSION

The present study focused on the synthesis, structural analysis, surface morphology evaluation, and electrochemical characterization of Ag/SmFeO<sub>3</sub> nanostructures prepared through a hydrothermal synthesis route. The main objectives of this work was to develop a rare-earth modified perovskite nanomaterial with controlled crystallinity, nanoscale particle size, and improved electrochemical properties suitable for advanced material and sensing applications. The hydrothermal method was selected because it provides a controlled environment for nucleation and crystal growth, enabling the formation of well-defined nanostructures with high crystallinity.

Ag/SmFeO<sub>3</sub> nanoparticles were successfully synthesized using high-purity precursor materials containing silver (Ag), samarium (Sm), and iron (Fe). During the synthesis process, urea was used as stabilizing and structure-directing agents to ensure homogeneous mixing of metal ions and to control particle growth. The hydrothermal reaction was conducted at 180 °C for 12 hours in a Teflon-lined stainless-steel autoclave. Under these controlled conditions, the formation of nanoscale perovskite structures was achieved through nucleation, growth, aggregation, and recrystallization processes.

The structural properties of the synthesized Ag/SmFeO<sub>3</sub> nanostructures were analyzed using X-ray diffraction (XRD). The XRD patterns confirmed the successful formation of the orthorhombic perovskite phase of SmFeO<sub>3</sub> along with the cubic phase of silver. The diffraction peaks matched well with the standard JCPDS files (Ag: 96-901-3046 and SmFeO<sub>3</sub>: 43-154-2658), confirming the crystalline nature of the synthesized materials. The absence of additional peaks related to secondary phases indicates that the prepared nanostructures possess high phase purity. Furthermore, the diffraction peaks were sharp and intense, which demonstrates the good crystallinity of the synthesized nanoparticles. Slight peak shifts and broadening observed in the XRD pattern indicate the successful incorporation of Ag and Sm ions into the Fe–O lattice structure. The average crystallite size was estimated using the Debye–Scherrer equation and was found to be approximately 54 nm. This nanoscale crystallite size indicates that the hydrothermal method effectively controls particle formation and prevents excessive grain growth. Smaller crystallite size significantly increases the surface-to-volume ratio, which is highly beneficial for catalytic and electrochemical applications.

The surface morphology of the synthesized Ag/SmFeO<sub>3</sub> nanostructures was investigated using scanning electron microscopy (SEM). The SEM images revealed the formation of rectangular nanostructures along with aggregated nanoparticles. The rectangular structures had an approximate size of about 500 nm and were formed through a self-assembly process during hydrothermal synthesis. Each rectangular structure was composed of multiple nanoscale particles assembled together. The morphology formation was influenced by various intrinsic and extrinsic factors, including reaction diffusion, crystal orientation, surface energy, and the presence of surfactants. Urea acted as a structure-directing surfactant that facilitated the adsorption of hydroxyl ions on the particle surface, guiding the growth of nanostructures along specific crystallographic directions. The formation mechanism involved nucleation, aggregation, and recrystallization processes followed by Ostwald ripening, which ultimately resulted in well-formed nanostructures with relatively uniform morphology. The observed porous and aggregated structure increases the available active surface area, which can enhance catalytic and electrochemical performance.

Electrochemical characterization of the Ag/SmFeO<sub>3</sub> nanostructures was performed using cyclic voltammetry (CV) and electrochemical impedance spectroscopy (EIS) to evaluate their electron transfer properties and electrochemical activity. CV studies were conducted using a glassy carbon electrode (GCE) modified with Ag/SmFeO<sub>3</sub> nanostructures in a ferrocyanide redox system. The CV results showed that the Ag/SmFeO<sub>3</sub>-modified electrode exhibited a significantly higher current response compared to the bare GCE. This enhancement indicates improved electron transfer kinetics and a larger electrochemically active surface area. The improved performance can be attributed to the synergistic interaction between silver nanoparticles and the SmFeO<sub>3</sub> perovskite structure. Silver enhances the electrical conductivity of the material, while the perovskite oxide structure provides excellent catalytic activity, resulting in faster electron transfer and improved electrochemical performance.

Further electrochemical evaluation was performed using electrochemical impedance spectroscopy. The EIS Nyquist plots revealed a significant reduction in charge transfer resistance for the Ag/SmFeO<sub>3</sub>-modified electrode compared to the bare GCE. The calculated charge transfer resistance ( $R_{ct}$ ) decreased from 824.6  $\Omega$  for the bare electrode to 366.2  $\Omega$  for the modified electrode. The smaller semicircle diameter observed in the Nyquist plot indicates enhanced electron transfer capability and improved conductivity of the nanostructured material. The porous morphology and high surface area of Ag/SmFeO<sub>3</sub> contribute to efficient electron transport pathways, which further improves electrochemical performance.

In conclusion, Ag/SmFeO<sub>3</sub> nanostructures were successfully synthesized through a hydrothermal method with controlled crystallinity, nanoscale particle size, and desirable morphological features. Structural analysis confirmed the formation of a highly crystalline and phase-pure perovskite structure, while SEM analysis demonstrated the formation of rectangular self-assembled nanostructures composed of nanoscale particles. Electrochemical studies revealed that Ag incorporation significantly improves conductivity, electron transfer kinetics, and electrochemical activity. The synergistic combination of silver nanoparticles and perovskite SmFeO<sub>3</sub> oxide results in enhanced electrochemical properties. Therefore, the synthesized Ag/SmFeO<sub>3</sub> nanostructures show strong potential as promising electrode materials for advanced electrochemical sensors, catalytic systems, and other energy-related applications.

## REFERENCES:

1. Poole, C. P., & Owens, F. J. (2003). *Introduction to Nanotechnology*. Hoboken, New Jersey: John Wiley & Sons
2. Kulkarni, S. K. (2015). *Nanotechnology: Principles and Practices* (3rd ed.). Cham, Switzerland: Springer.
3. Ratner, M. A., & Ratner, D. (2003). *Nanotechnology: A Gentle Introduction to the Next Big Idea*. Upper Saddle River, NJ: Prentice Hall.
4. Vollath, D. (2013). *Nanomaterials: An Introduction to Synthesis, Properties and Applications* (2nd ed.). Weinheim, Germany: Wiley-VCH.
5. Cao, G., & Wang, Y. (2011). *Nanostructures and Nanomaterials: Synthesis, Properties and Applications* (2nd ed.). Singapore: World Scientific Publishing.
6. Schmid, G., *Nanoparticles: From Theory to Application*, Wiley-VCH, 2010.
7. Hornyak, G. L., Tibbals, H. F., Dutta, J., & Rao, A. K., *Introduction to Nanoscience and Nanotechnology*, CRC Press, 2008.
8. Cao, G., & Wang, Y., *Nanostructures and Nanomaterials: Synthesis, Properties and Applications*, World Scientific, 2011.
9. Feynman, R. P., "There's Plenty of Room at the Bottom," American Physical Society Meeting, 1959.
10. Wilson, M., Kannangara, K., Smith, G., Simmons, M., & Raguse, B., *Nanoscience and Nanotechnology: Fundamentals to Frontiers*, CRC Press, 2002.
11. Cao, G., & Wang, Y., *Nanostructures and Nanomaterials: Synthesis, Properties and Applications*, World Scientific, 2011.
12. Efros, A. L., Lockwood, D. J., & Tsybeskov, L., *Semiconductor Nanocrystals: From Basic Principles to Applications*, Springer, 2003.
13. Kreibig, U., & Vollmer, M., *Optical Properties of Metal Clusters*, Springer, 1995.
14. Cullity, B. D., & Graham, C. D. (2009). *Introduction to Magnetic Materials* (2nd ed.). Hoboken, NJ: John Wiley & Sons.
15. Hill, T. L. (1994). *Thermodynamics of Small Systems*. New York: Dover Publications.
16. Dowling, N. E. (2013). *Mechanical Behavior of Materials* (4th ed.). Boston, MA: Pearson.
17. Heiz, U., & Landman, U. (2007). *Nanocatalysis*. Berlin: Springer.
18. Edelstein, A. S., & Cammarata, R. C. (1998). *Nanomaterials: Synthesis, Properties and Applications*. Bristol: Institute of Physics Publishing.

19. Nanoparticle Technology Handbook edited by Makio Naito, Masuo Hosokawa, Toyooki Yokoyama, and Kiyoshi Nogi